

Dr. Parth Nayak

AI Researcher | Data Scientist | Astrophysicist

📍 Munich, Germany

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EDUCATION

🎓 **Ph.D. in AI & Astrophysics**

LMU Munich | Jan 2026

🎓 **B.Sc.+M.Sc. in Physics** (dual degree)

IISER Bhopal | Jul 2020

PUBLICATIONS

- Nayak et al. 2025, [arXiv:2510.19899](#)
- Chang, Nayak et al. 2025, [arXiv:2508.03264](#)
- Nayak et al. 2024, [A&A 689-A153](#)
- Nayak & Saha 2022, [MNRAS 510-2](#)

TECHNICAL SKILLS

ML / Deep Learning

Neural networks (CNNs, MLPs), simulation-based inference (SBI), generative models (normalizing flows, diffusion models, autoencoders, GANs)

Other AI / Statistics

LLM fine-tuning & prompt engineering, LLM output evaluation & ranking (RLHF), evolutionary algorithms, Gaussian processes, Bayesian & Monte Carlo methods, mixture models

Programming

Python (proficient); HTML, CSS, Javascript (intermediate); C++, Java (basic)

Python packages

TensorFlow, Keras, PyTorch (familiar), Optuna, NumPy, SciPy, Pandas, Matplotlib, H5Py, AstroPy, HealPy, among others

Other IT

High-performance computing (HPC), Git version control, LaTeX typesetting, Markdown, Mac/MS Office, Shell (bash, Z shell)

LANGUAGES

English (full working proficiency); Gujarati, Hindi (native/bilingual); German (B1/intermediate); French (beginner)

HIGHLIGHTS

- 5+ years of research experience in Data Science and AI algorithm development
- Authored 4 research papers on novel AI applications in astrophysics
- Delivered 10+ technical talks at international conferences and research institutes
- Received 5+ academic awards and fellowships

RELEVANT EXPERIENCE

Doctoral Candidate: AI & Astrophysics LMU Munich | Oct 2021 - Jan 2026

Hightech Agenda Bayern chair of Astrophysics, Cosmology & AI + Munich Center for Machine Learning (MCML)

- Developed a ML inference framework “*LyaNNA*” ([🔗 lyanna](#)) based on a novel 1D ResNet with Python/TensorFlow from scratch that outperforms conventional methods by >100% in inference precision for complex, realistic, noisy spectroscopic data (simulation-based inference)
- Developed “*SynTH*” ([🔗 synth](#)) to model intergalactic absorption from massive cosmological hydrodynamic simulations and synthesize training data for *LyaNNA*
- Authored 3 research papers on AI for inference in astrophysics [[X Nayak+ 2024](#), [2025](#), [Chang+ 2025](#)]

Master's Thesis: AI & Astrophysics IISER Bhopal | Aug 2019 - Jul 2020

- Developed a Genetic Algorithm framework “*GA-ILC*” to optimize the reconstruction of cosmic microwave background (CMB) from contaminated observations
 - ▶ A novel, scalable evolutionary algorithm ([🔗 geneticalpy](#)) to extract faint cosmological signals from high-noise environments (foreground removal)
- Authored one paper on evolutionary algorithms in astrophysics [[X Nayak & Saha 2022](#)]

Freelance Data Annotator DataAnnotation | May - Jun 2026

- Ranked and evaluated LLM outputs and assessed prompt-response quality on a contract basis for model training pipelines

LEADERSHIP & SUPERVISION

- Co-supervisor of LMU Master's theses and labs Oct 2021 - Dec 2025
Designed and supervised two LMU Master's theses, and co-supervised 5+ computational and instrumental lab courses
- Co-organizer of “Astronomy on Tap” (AoT) Munich since Oct 2025
Coordinating the monthly outreach event to engage the general public with accessible scientific talks and pub-quizzes
- Co-organizer of “Meet the Speaker” @LMU Observatory Oct 2023 - May 2025
Revitalized the weekly series by relocating it to the historic 200-year old telescope dome, a unique atmosphere for students for informal networking with international experts
- Coordinator & Senior Advisor of IISERB Astronomy Club (IBAC) 2016 - 2020
Played a foundational role as the club's second coordinator, establishing enduring traditions and growing the community into one of India's most popular student astronomy clubs
- Physics Department Representative Aug 2017 - Jul 2018
Unanimously elected to serve as the primary liaison between faculty and a class of 60+ physics major students

SELECTED AWARDS & FELLOWSHIPS

- DAAD Research Grant — Ph.D. scholarship Germany | Oct 2021 - Dec 2025
- Proficiency Medal in Physics — IISER Bhopal India | Jul 2020
- DAAD-WISE Scholarship Germany | May - Jul 2019
- INSPIRE Scholarship for Higher Education India | Aug 2015 - Jul 2020